BonPlan.ai
Tell us When. We Tell the How.

The Vision

1.0 Execution Summary

BonPlan.ai is the world's first AI travel agent built entirely on **Constraint-Based Planning**. Tell us your dates and destinations, and our agent handles the complex logistics. Unlike standard travel generators that build rigid lists, BonPlan.ai is a dynamic engine designed to respect your non-negotiables, adapt on the fly, and act as your end-to-end companion from planning to departure.

1.1 Core Capabilities

- ### 1.1.0 Smart Anchors (Constraint-Based Planning)

  Feed the AI your pre-booked flights, non-negotiable dinner reservations, or scheduled tours. BonPlan locks these anchors in place and intelligently routes the rest of your trip around them, ensuring zero logistical conflicts.

- ### 1.1.1 Customized Pacing

  Whether you want an aggressively efficient sightseeing tour, a relaxed vacation, or a mix of both, the AI optimizes the travel routes and density of activities based on your exact desired pace.

- ### 1.1.2 Smart Adjustments (Dynamic Auto-Shuffling)

  Travel plans change. Using an intuitive drag-and-drop timeline, users can extend or move an activity, and the AI will instantly reshuffle the rest of the day to make the new schedule fit seamlessly.

- ### 1.1.3 Specific Regenerations

  Don't like a specific museum suggestion? Highlight a single activity and ask the AI to swap it out for a food tour, leaving the rest of your meticulously planned day completely untouched.

- ### 1.1.4 Multiplayer Collaborative Mode

  Planning a group trip is no longer a headache. Invite friends or family to the itinerary, let them vote on generated activities, and watch the AI automatically adapt the schedule based on group consensus.

- ### 1.1.5 Conversational Context ("Trip For All")

  Speak to BonPlan exactly how you would speak to a human travel agent. Just describe your vibe—whether it's a solo backpacking trek, a family-friendly holiday, or an adventurous weekend with friends—and the AI tailors the recommendations to perfectly match the context.

- ### 1.1.6 The Real-Time Travel Concierge

  BonPlan doesn't stop working when the itinerary is finalized. Throughout your journey, the agent acts as your live companion, providing booking updates, next-day reminders, localized weather alerts, traffic routing, and instant contingency plans for last-minute changes.

2.0 Problem Statement & Market Gap

Modern travel planning is a logistics nightmare. The average traveler juggles dozens of open tabs across flights, hotels, maps, review sites, and spreadsheets just to build a single itinerary. If one element changes—a delayed flight, a closed museum, or a newly discovered restaurant—the entire house of cards collapses, forcing the user to manually recalculate travel times and reshuffle their entire day.

While current AI travel tools (like ChatGPT, Roam Around, or basic trip generators) can easily create a generic "5 days in Paris" list from a blank slate, they completely fail at the reality of how humans actually travel. Real trips are built around hard constraints. When a traveler tries to tell a standard AI, "I have a 14-hour international flight landing at 3:00 PM, and a non-refundable dinner reservation at 8:00 PM—plan the hours between," current tools break down. They suffer from Context Amnesia - They overwrite or delete the user's hard-coded reservations when regenerating a day. They suggest activities that are physically impossible to travel between in the allotted time. They fail to account for the real-world friction of transit, baggage claim, or check-in times.

2.1 The Solution

Travelers don't want an AI to dictate their entire trip. They want an AI to handle the tedious logistical math between the things they already know they want to do. BonPlan.ai exists to solve the "routing and fitting" problem. By treating user inputs as immutable **"Smart Anchors"** BonPlan shifts AI travel from a static text generation exercise into a dynamic, mathematical scheduling engine. And yes, it can still generate static end-to-end trips.

2.2 Technological Viability

A year ago, building this was impossible because Large Language Models were strictly generative—they predicted the next word. Today, with the rise of Agentic AI, we can give models access to external tools (like live maps, routing algorithms, and flight APIs). BonPlan.ai leverages this exact shift. It isn't just generating text, it is an autonomous agent reasoning through geographical constraints, calling APIs to verify distances, and building a flawless schedule around the user's non-negotiable anchors.

3.0 Target Personas

3.0.0 The "Type-A" Planner (The Anchor Heavy User)

Who they are: The meticulous traveler who already has a massive spreadsheet. They book Michelin-star restaurants months in advance and have their long-haul flights (like a 16-hour SFO to DEL route) locked in tightly.
Their Pain Point: They hate current AI tools because standard generative AI always overwrites their prized reservations or fails to account for jet lag and transit times.
How BonPlan solves it: Smart Anchors. They can hardcode their flights and dinner reservations, knowing the AI will respect those boundaries like a brick wall and flawlessly calculate the transit time between their hotel and their locked-in plans.

3.0.1 The Group Trip "Cat Herder"

Who they are: The designated organizer for a friend group, a hackathon squad, or a complex family vacation involving multiple generations (like coordinating activities that work for both adults and a young nephew).
Their Pain Point: Juggling endless group chat messages, conflicting budgets, and differing opinions. Someone wants to go hiking, someone else wants to relax, and building a consensus is a nightmare.
How BonPlan solves it: Multiplayer Collaborative Mode. They simply invite the group to the canvas. The AI generates options, the group votes, and the itinerary dynamically updates based on the winning choices. It completely removes the friction of group decision-making.

3.0.2 The Dynamic Explorer (The Solo/Hobbyist Traveler)

Who they are: A traveler focused on specific experiences—like capturing the perfect golden hour photography, hitting specific hiking trails, or maintaining their gym routine while abroad. They have a general idea of what they want but need to adapt on the fly.
Their Pain Point: Weather changes, or a specific location is too crowded, and their rigid PDF itinerary is suddenly useless.
How BonPlan solves it: Specific Regenerations & Smart Adjustments. If the weather ruins a planned outdoor shoot, they can highlight that single block of time and hit "Swap for an indoor activity." The AI updates that single slot without destroying the rest of the day's flow.

3.0.3 The "Bleisure" Traveler (Business + Leisure)

Who they are: Someone traveling primarily for work, a conference, or a university event, but extending the trip by a few days to explore.
Their Pain Point: They have massive, immovable blocks of time in the middle of their day (e.g., "Conference from 9 AM to 4 PM") and need to know exactly what is feasible to do in the 3-hour gap before dinner.
How BonPlan solves it: Constraint-Based Gap Filling. They block out their work hours as anchors. The AI acts as a hyper-efficient assistant, looking at the map and finding the highest-value sightseeing or dining options perfectly clustered near their conference center.

3.0.4 The "Blank Canvas" Traveler (The Overwhelmed Beginner)

Who they are: They managed to book a flight to a destination they wanted to go to because they had a week off, but that is exactly where their planning stopped. They know nothing about the city's layout, the must-see landmarks, or the local culture. They are starting from absolute zero.

Their Pain Point: Severe decision fatigue. Staring at a blank itinerary is paralyzing, and researching means reading hundreds of conflicting travel blogs, watching endless TikToks, and trying to figure out if a 3-day pass to a museum is actually worth it.

How BonPlan solves it: "Trip For All" Conversational Context & End-to-End Companion. They don't need to use Smart Anchors right away. They simply type: "I am going to Tokyo for 5 days. I know nothing. Make it a mix of big tourist spots and chill evenings." BonPlan instantly generates a flawless, geographically logical baseline itinerary. Once they land, the app's real-time concierge acts as their personal guide, telling them exactly how to get to the next spot and what the weather will be like, completely removing the anxiety of navigating a new place.

4.0 The High Level Architecture

4.0.0 The Brain & The Body (The Tech Stack)

To build an AI that can actually reason through real-world logistics, we are using a modern, hyper-fast stack built for autonomous agents.

The Interface: A highly interactive, drag-and-drop frontend calendar that updates in real time.

The Engine: An asynchronous, AI-first backend that orchestrates the complex logic and background API calls without freezing the user experience.

The Memory: A strict relational database that locks in your "Anchors" so the AI knows exactly what it is legally allowed to touch and what is strictly off-limits.

The Orchestrator: An agentic framework that acts as the "brain," giving the LLM the ability to think step-by-step and decide which external tools it needs to solve your specific travel puzzle.

4.0.1 The Constraint-Solving Engine (The Agentic Loop)

When a user drops in for a 2:00 PM flight and an 8:00 PM dinner, the AI doesn't just guess what goes in between. It follows a strict, logical pipeline.

Extraction: It identifies and locks in the non-negotiable anchors.

Calculation: It does the math to find the exact "whitespace" (free time) available, factoring in realistic transit and check-in times.
Verification: The AI reaches out to real-world maps and databases to find activities that physically fit into that specific timeframe.
Generation: Once the math is proven, it perfectly places the new activities onto the user's visual calendar.

4.0.2 Surgical Adjustments (The Swap Feature)

If a user wants to change a single museum visit, the system doesn't waste time and resources regenerating the entire trip. It performs a surgical swap. The AI isolates that specific two-hour window, checks the geographical distance to the surrounding events, and patches in a new recommendation that fits perfectly without toppling the rest of the day's dominoes.

4.0.3 Real-World Grounding (External Integrations)

An AI is only as smart as the reality it understands. BonPlan is directly wired into the real world via external APIs to ensure every itinerary is actually executable.
Live Maps & Places: To calculate exact walking/driving/transit times and verify that a restaurant is actually open on a Tuesday.
Live Travel Data: To pull real flight schedules and hotel availability.
Dynamic Weather: To proactively adjust plans (e.g., swapping a park walk for an indoor gallery if rain is suddenly forecasted).

5.0 Risks

5.0.0 The "Domino Effect" Algorithm

The Unknown: If a user drags a 1-hour museum visit and extends it to 3 hours, how exactly do I calculate the ripple effect?
The Question: How do I write the backend logic that pushes every subsequent activity ahead by 2 hours, recalculates the transit times, and then stops the domino effect right before it hits a "Smart Anchor" (like the 8 PM dinner)? Do I drop an activity, or just shorten them all?

5.0.1 Managing API Latency & UX

The Unknown: Agentic loops are inherently slow. If the LLM has to call the Google Places API, then the Maps API, then reason about the data, it could take 15–30 seconds to generate a day.
The Question: How do I keep the user engaged on the React frontend while the FastAPI backend is running this heavy orchestration? Do I stream the agent's "thoughts" to the UI (e.g., "Looking for restaurants near the Eiffel Tower...") so it feels fast?

5.0.2 Hallucination Guardrails

The Unknown: LLMs are notorious for being confidently wrong.
The Question: Even if I hook the agent up to the Google Places API, what happens if the API returns incomplete data? How do I strictly enforce that the LLM cannot schedule a visit to a landmark on a Tuesday if it couldn't verify the opening hours?

5.0.3 Token Cost Optimization

The Unknown: An autonomous agent that makes multiple tool calls per user query can burn through API tokens incredibly fast.
The Question: How do I prevent the agent from getting stuck in an infinite loop? (e.g., If it searches for a specific type of restaurant in a 10-minute gap, fails, and just keeps retrying). How do I set a "max_iterations" limit so a single complex query doesn't cost a fortune?

5.0.4 Handling Infeasible Constraints

The Unknown: Users will inevitably ask for the impossible. (e.g., "I land at JFK at 4 PM, and my Smart Anchor is a Broadway show at 5 PM.")
The Question: How does the agent elegantly push back? Does it just throw an error, or does the prompt architecture need a specific "Negotiation Mode" where it tells the user exactly why the math doesn't work and suggests a compromise?

5.0.5 State Syncing with Drag-and-Drop

The Unknown: When a user is rapidly moving items around on a visual calendar, the frontend state changes instantly.

The Question: How do I prevent race conditions? If they swap two events on the frontend, do I lock the UI with a spinner until the backend recalculates the new transit times, or do I do optimistic UI updates and risk the math being wrong for a few seconds?